MILITARY SPECIFICATION

ARMING/FIRING UNIT

This specification is approved for use by US Army Armament Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of arming/firing unit (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-454	-	Standard General Requirements for Electric Equipment.
MIL-STD-1188	-	Commercial Packaging of Supplies and Equipment.

DRAWINGS AND TECHNICAL DATA PACKAGE LIST (TDPL)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

TDPL 13-12-111 - Arming/Firing Unit.

FSC 1040

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-TSC-S, Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on TDPL 13-12-111 shall conform to the specifications listed thereon or to the specific characteristics set forth on the drawings.

3.1.2 Components. All components of the arming/firing unit shall conform to the specifications and drawings listed on TDPL 13-12-111.

3.2 Assembly. The arming/firing unit shall be assembled in accordance with Drawing E13-12-111.

3.2.1 Insulation resistance. The arming/firing unit shall have an insulation resistance of not less than 10 megohms when tested as specified in 4.4.4.3.

3.3 Continuity. The continuity of the arming/firing unit shall comply with the requirements of table I when tested as specified in 4.4.4.1.

3.4 Leakage. The arming/firing unit shall show no evidence of leakage when tested as specified in 4.4.4.2.

3.5 Preproduction. Prior to the start of regular production, a preproduction sample of arming/firing units shall be produced (see 6.2) in accordance with this specification for examination and tests (see 4.3).

3.6 Workmanship. The arming/firing unit shall be free from contamination (oil, rust, grease or foreign matter) and damage such as dents, cracks or deformation which could affect its intended use and the arming/firing unit shall comply with the applicable requirements of MIL-STD-454 for workmanship.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Contractor's responsibility. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

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TABLE I. Continuity requirements - electrical resistance

Pin No	Pin No	ON/OFF arming switch position: Arm switch OFF	Arm switch ON Fire switch not dep	Arm switch ON Fire switch depressed
A to	B	#	Lamp resistance	Lamp resistance
	C	#	#	0
	D	#	Lamp resistance	Lamp resistance
	E	#	#	0
	F	#	Lamp resistance	Lamp resistance
B to	C	#	#	Lamp resistance
	D	0	0	0
	E	#	#	Lamp resistance
	F	0	0	0
C	D	#	#	Lamp resistance
	E	0	0	0
	F	#	#	Lamp resistance
D	E	#	#	Lamp resistance
	F	0	0	0
E	F	#	#	Lamp resistance

NOTES: 1. Letters A thru F are connector pin identification
2. 0 indicates zero resistance
indicates infinite resistance
3. Lamp resistance - indicator light resistance varies from one light to another and therefore should be removed and checked prior to testing of unit. Normal resistance is 150 to 250 ohms.

4.1.2 Objective evidence. The contractor shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Classification of inspections.

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 Preproduction inspection.

4.3.1 Sample. A preproduction sample of 5 arming/firing units shall be produced using the same methods, materials, equipment and processes as will be used in regular production.

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4.3.2 Inspection procedure.

4.3.2.1 For examination and test. The sample arming/firing units shall be examined and tested in accordance with the classification of defects (4.4.3.2). The AQLs do not apply.

4.3.3 Acceptance/rejection criteria. The preproduction sample shall comply with the examinations and tests specified in 4.3.2 to be acceptable. The contractor shall obtain written approval from the contracting officer prior to proceeding with regular production.

4.4 Quality conformance inspection.

4.4.1 Lotting. A lot shall consist of the units offered for acceptance at one time which have been produced by one manufacturer, at one plant, under essentially the same manufacturing conditions and in essentially the same time frame.

4.4.2 Sampling for examination and tests. Sampling of the units shall be conducted in accordance with MIL-STD-105.

4.4.3 Inspection procedure.

4.4.3.1 For examination and tests. The sample units and the level A preparation for delivery shall be examined and tested in accordance with the classification of defects and MIL-STD-105.

4.4.3.2 For critical defects. Each item in the lot shall be tested for the critical characteristics listed in the classification of defects.

4.4.3.3 Classification of defects. The AQL's apply to the group of defects unless otherwise specified.

(a) Arming/firing unit (Drawing E13-12-111).

Categories	Defects	Acceptance standards
Critical:		
1	Continuity incorrect	4.4.4.1
Major:	AQL 1.0 percent defective	
101	Name plate locations or application incorrect	
102	Insulation resistance	4.4.4.3
103	Threads damaged	
104	Screws loose or missing	
105	Protective finish missing or incorrect	
106	Workmanship (3.6)	

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Minor: AQL 1.5 percent defective

201	Leakage	4.4.4.2

(b) Name plates (Drawing C13-12-118).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Dimensions incorrect
102	Material incorrect
103	Printing incorrect

(c) Packaging (section 5).

Categories	Defects
Critical:	None defined
Major	AQL 4.0 percent defective
101	Marking illegible or incorrect
102	Packaging material incorrect
103	Dimensions incorrect
104	Item loose in pack
105	Barrier bag not closed properly
106	Box closure not in accordance with applicable box specification

4.4.4 Tests. The tests shall be performed as follows:

4.4.4.1 Continuity. Using an ohmmeter determine resistance between each pin of connector. Resistance shall be as indicated in table I to be acceptable.

4.4.4.2 Leakage. The arming/firing unit shall be submerged in 3.0 ± 0.5 feet of water (914.40 millimeters) at ambient temperature for 15 minutes. Upon removal, the unit shall be thoroughly dried and the base plate removed. Any water found inside the unit constitutes leakage.

4.4.4.3 Insulation resistance. The insulation resistance between pairs of component circuits and between each circuit and ground shall be as specified in 3.5 when measured at 500 volt direct current (vdc) minimum, for a minimum of 30 seconds.

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5. PACKAGING

5.1 Preservation. Preservation of the arming/firing unit shall be in accordance with Drawing D13-12-127.

5.2 Packing. Packing of the arming/firing unit shall be in accordance with the applicable requirements of MIL-STD-1188.

5.3 Marking. Marking of the preserved unit shall be in accordance with Drawing D13-12-127. Marking of the packed unit shall be in accordance with the applicable requirements of MIL-STD-1188.

6. NOTES

6.1 Intended use. This item is a component of the M243 and M259 smoke grenade launchers.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Preproduction.

(1) Time allowed for contractor submission of samples for Government test and evaluation after award of contract.
(2) Name and address of test facility and shipping instructions when testing is performed by the Government.
(3) Time required for the Government to notify the contractor whether or not to proceed with production.

6.3 Supersession data. This military specification supersedes Chemical Systems Laboratory Purchase Description EA-A-1182 dated 27 April 1979.

Custodian:

Army - EA

Preparing activity:

Army - EA

Project No. 1040-A150

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DOCUMENT IDENTIFIER *(Number)* AND TITLE
MIL-A-51487(EA) Arming/Firing Unit

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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